S. mithesh
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SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I S. mithesh *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

level - 3

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Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	30%	Demonstrates thorough understandin g; answers all questions accurately	Shows good understandin g with few errors	Partial understandin g; several incorrect responses	Lacks understandin g; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterpret s a few questions	Often misinterprets questions	Unable to understand questions

AT - 1

Artificial Intelligence:

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1. Bidecrectional search vs Breadth - First search (BFS)

- Breadth - First Search (BFS): Explores nodes layer -by- layer starting from node S until it Reaches a single goal.
- Bidrectional search: Runs two simultaneous BFS traversals - one forward from node S and one backward from node G, the search terminates as soon as the two frontiers intersect.

Time complexity Analyses

For a uniform branching factor b and shortest Path distance d:

- BFS: generates $O(b^d)$ nodes.
- Bidirectional search: Generates two smaller Trees of depth $d/2$, Resulting in $O(b^{d/2} + b^{d/2})$ $= O(b^{d/2})$ complexity.

Eg: 10 node graph,
$b = 3$, $d = 4$:
BFS: Expands down to level 4
$\approx 1 + 3 + 9 + 27 + 81 = 121$ nodes

Bidirectional: Both go to one level
$2 \to (1 + 3 + 9) \times 2 = 26$ nodes

2. Agent Formulation for a self-Driving car

ans: Percepts and Actions

Percepts: High-resolution vedio feeds (cameras), paint cloud (LIDAR/Radar), spatial positions (GPS), speed/acceleration).

Actions: streering angle (left, right, straight), acceleration (throttle control), braking gear shifting, signaling (turn indicators, hazard, lights)

Environmental ~~changes~~ characterization:

- Partially observable: sensor range is limited by visual oclusions (eg. blind spots, heavy, rain, buildings and the intent/states of other drivers cannot be ~~too~~ directly observed.

- Stochastic: the consequences of actions are non-deterministic due to upredictable road adhesion, weather shifts mechanical variance and non-deterministic behaviour of other drivers or pedestrians.

- Dynamic: the environment changes continuosly while the agent deliberates; Pedestrians move, Traffic lights change, and incoming traffic advancences uniform cost Search (UCS) trace

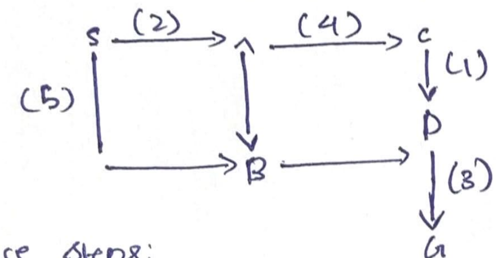
graph setup

consider a ride-hailing road network with 6 nodes.
(S, A, B, C, D, G):

- Nodes: S (start), A, B, C, D, G (goal)
- weighted Edges (Travel cost = time + Fuel):

~~For~~

(S,A) = 2, (S,B) = 5 (A,B) = 1,
(A,C) = 4, (B,D) = 2, (C,D) = 1
(C,G) = 6, (D,G) = 3



Trace steps:

1. Pop S (g=0): Explored {S}
   Frontier: [(A, g=2, S→A), (B, g=5, S→B)]
2. Pop A (g=2): Explored {S, A}
   Frontier: [(D, g=5, S→A→B→D), (C, g=6), S→A→C)
3. Pop D (g=5): Explored: {S, A, B, D}
   Frontier: [(C, g=6, S→A→C), (G, g=8, S→A→B→D→G)]

(~~6. pop G (g=8):~~ ~~goal Reached. search terminates.~~

5 Pop C (g=6): Explored {S, A, B, D, C}
   Frontier: [(G, g=8, S→A→B→D→G)]
6. Pop G (g=8): goal reached search terminates

$\boxed{\text{Total cost: 8}}$